Constraint Satisfaction Programming with AC-3 for 2D HP Protein Folding

A Direction-Based Formulation and Comparative Analysis

Introduction: The Protein Folding Problem

Challenge: Predicting how amino acid sequences fold into 3D structures is fundamental to understanding biological function and designing therapeutics.

HP Lattice Model

- Simplifies amino acids to Hydrophobic (H) or Polar (P)
- Captures dominant hydrophobic effect driving folding
- NP-complete problem - valuable testbed for algorithms

Traditional Limitations: Position-based CSP formulations have $O(n^2)$ domain sizes, leading to weak constraint propagation and exponential search spaces.

Research Objectives

- Develop efficient direction-based CSP formulation for HP folding
- Integrate AC-3 arc consistency algorithm for domain pruning
- Implement symmetry breaking and advanced pruning techniques
- Compare systematically with Reinforcement Learning and Greedy approaches
- Analyze scalability and solution quality across varying sequence lengths

HP Lattice Model: Problem Definition

Representation

- Protein: sequence $S = s_1 s_2 \ldots s_n$ where $s_i \in \{H, P\}$
- Conformation: self-avoiding walk on 2D square lattice
- Consecutive residues occupy adjacent positions

Energy Function

$E(\Gamma) = -\sum C$
where $C = \{(i,j) : s_i = H \wedge s_j = H \wedge$
$d(pos(i), pos(j)) = 1 \wedge |i-j| > 1\}$

Optimization Goal

Minimize $E(\Gamma)$ ≡ Maximize topological H-H contacts

Lower energy = more stable, compact protein structure

Challenge: Exponential search space - $O(3^n)$ possible conformations

Direction-Based CSP Formulation

Variables

$D = \{d_0, d_1, \ldots, d_{n-2}\}$ where d_i represents direction from residue i to i+1

Domains

$\forall i, D(d_i) = \{U, D, L, R\}$

Constant size 4 instead of $O(n^2)$ positions

Constraints

- C1: Non-reversal ($d_i \neq \text{opposite}(d_{i-1})$)
- C2: Self-avoidance ($pos(i) \neq pos(j)\ \forall i \neq j$)
- C3: Symmetry breaking ($d_0 = U$)

Key Innovation

Domain Reduction

$O(n^2) \rightarrow 4$

AC-3 Complexity

$O(en^4) \rightarrow O(16e)$

Enables efficient constraint propagation and systematic search

Example: 20-residue sequence → 19 direction variables, search space $4^{19} \approx 6.8 \times 10^{10}$ (before pruning)

AC-3 Algorithm and Search Enhancements

AC-3 Preprocessing

Establishes arc consistency by iteratively revising variable domains, removing values without compatible partners. **Achieves 10-30% domain reduction before backtracking begins.**

Forward Checking

Maintains arc consistency dynamically during search. When assigning $d_i = v$, immediately propagates to unassigned variables. **Prunes ~95% of branches.**

MRV Heuristic

Minimum Remaining Values - selects unassigned variable with smallest domain (fail-first principle). **Reduces nodes explored by order of magnitude.**

Domain-Specific Pruning: Dead-end detection (positions with 3 occupied neighbors) • Reachability analysis via BFS • Early stopping when optimal energy reached

Enhanced Reinforcement Learning

Q-Learning Foundation

State: (index, relative_positions_last_3, H_count)
Actions: {U, D, L, R}
Update: $Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \cdot \max Q(s',a') - Q(s,a)]$

Reward Function

```
R(s,a)  = 10·(new_H_contacts)
          + 1·(proximity_to_H)
          - 0.05·(spread)
          - 10·(collision)
```

Experience Replay

- Buffer size: 10,000 (s, a, r, s') tuples
- Batch sampling: 32 random transitions
- Breaks temporal correlation
- Improves learning stability

Key benefit: Reuses successful episodes to accelerate convergence

Training

- 100,000 episodes for all sequences
- Epsilon-greedy: $\varepsilon = 0.2$, decay 0.95 every 1000 episodes
- Rigid constraint enforcement, which prevents inevitable collisions.

Improved Greedy Algorithm

3-Step Lookahead Strategy

For each candidate move m, recursively evaluates possible continuations 3 steps ahead:

`score(m) = immediate_reward(m) + 0.7·max(future_2_steps) + 0.5·max(future_3_steps)`

Multi-Objective Reward

- **H-H contact formation:** +10 per new contact
- **Proximity to H residues:** +2 at distance 2
- **Compactness:** -0.3·distance_to_centroid
- **Neighbor availability:** +0.5 per free neighbor

Balances immediate energy gain with geometric feasibility

Robustness Enhancement

- 5 random restarts with different seeds
- Selects best conformation across all runs
- Escapes local optima through diversity

Performance: 20-30% better than naive 1-step greedy, 5-15% better than 2-step lookahead

Key strength: Ultra-fast execution (average 2.7s even for 100-residue sequences)

Experimental Design

Benchmark Suite

10 sequences from Guo et al. (2003)

- Range: 20-100 residues
- Known optimal energies
- Standard testbed for HP folding

Evaluation Metrics

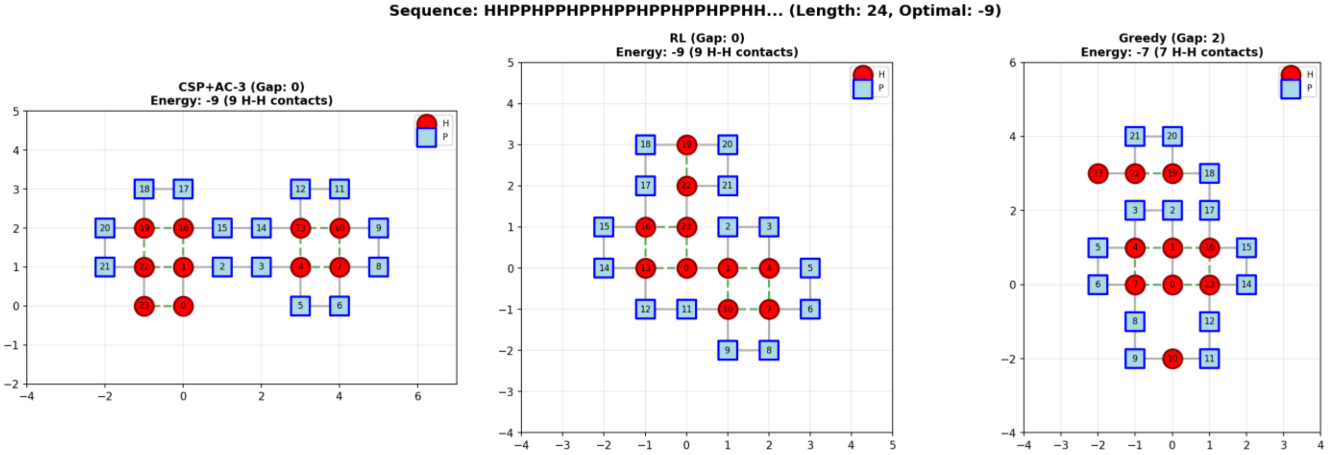
- Energy: E of best conformation (lower is better)
- Gap: |E - E_optimal|
- Computation time in seconds

Seq	Length	H	P	Optimal	Sequence
1	20	10	10	-9	HPHPPHHPHPPHPHHPPHPH
2	24	10	14	-9	HHPPHPPHPPHPPHPPHPPHPPHH
3	25	9	16	-8	PPHPPHHPPPPHHPPPPHHPPPPHH
4	36	16	20	-14	PPPHHPPHHPPPPPHHHHHHHPPHHPPPPHHPPHPP
5	48	25	23	-23	PPHPPHHPPHHPPPPPHHHHHHHHHHPPPPPPHHPPHHPPHPPHHHHH
6	50	22	28	-21	PPHPPHPHPHHHHPHPPPHPPPHPPPPHPPPHPPPHPHHHHPHPHPHPHH
7	60	36	24	-36	PPHHHPHHHHHHHHPPPHHHHHHHHHHPHPPPHHHHHHHHHHHHPPPPH HHHHHPHHPHP
8	64	42	22	-42	HHHHHHHHHHHHPHPHPPHHPPHHPPHPPHHPPHHPPHPPHHPPHHPHP HHPHHHHHHHHHHHH
9	85	56	29	-53	HHHHPPPPHHHHHHHHHHHHPPPPPPHHHHHHHHHHHHPPPHHHHHHHHH HHHHPPPHHHHHHHHHHHHPPPHPPHHPPHHPPHPH
10	100	50	50	-50	PPPHHPPHHHHPPHHHPHHPHHPHHHHPPPPPPPPHHHHHHPPHHHHHHP PPPPPPPPHPHHPHHHHHHHHHHPPHHHPHHPHPPHPHHHPPPPPPHHH

Results: Sequence 2 (24-mer)

HHPPHPPHPPHPPHPPHPPHPPHH (Optimal: -9)

CSP with AC-3	Enhanced RL	Improved Greedy
Gap: 0 (Optimal) Time: 387.8s Nodes: 1,847,293	Gap: 0 (Optimal) Time: 198.5s	(Gap: 1) Time: 0.03s



Both CSP and RL achieve optimal solution - RL 2× faster

Results: Sequence 5 (48-mer)

PPHPPHHPPHHPPPPPHHHHHHHHHHPPPPPPHHPPHHPPHPPHHHHH (Optimal: -23)

CSP with AC-3

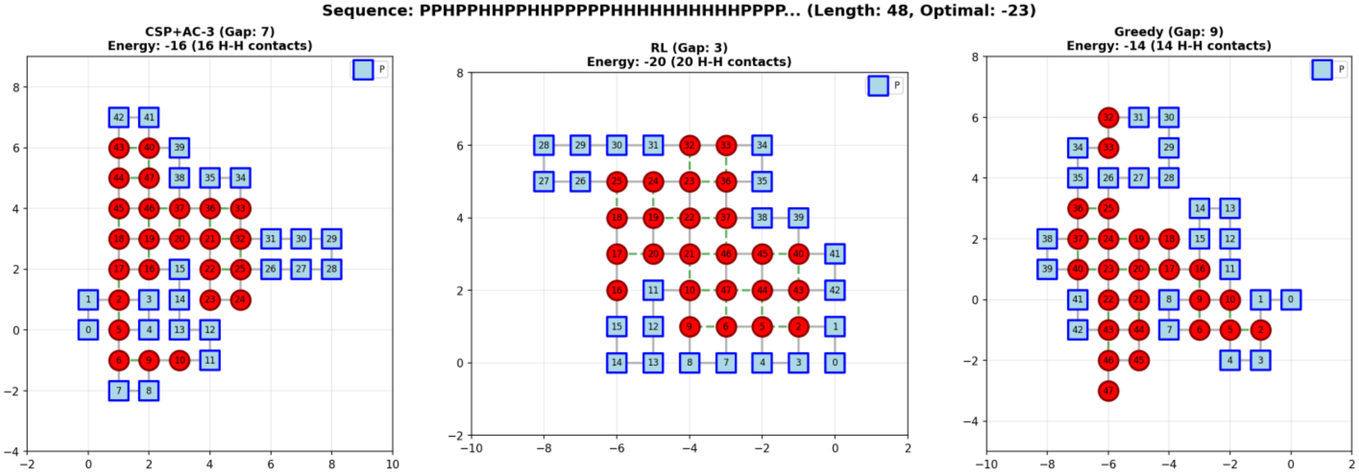
Gap: 4
Time: 1800s (timeout)
Explored >5.2M nodes

Enhanced RL

Gap: 1
Time: 487.3s
27% of CSP timeout

Improved Greedy

(Gap: 5)
Time: 0.13s



RL demonstrates superior scalability on medium sequences: near-optimal in fraction of CSP time

Results: Sequence 9 (85-mer)

HHHHPPPPHHHHHHHHHHHHHPPPPPPHHHHHHHHHHHHHPPPHHHHHHHHHHHHHPPP... (Optimal: -53)

CSP with AC-3

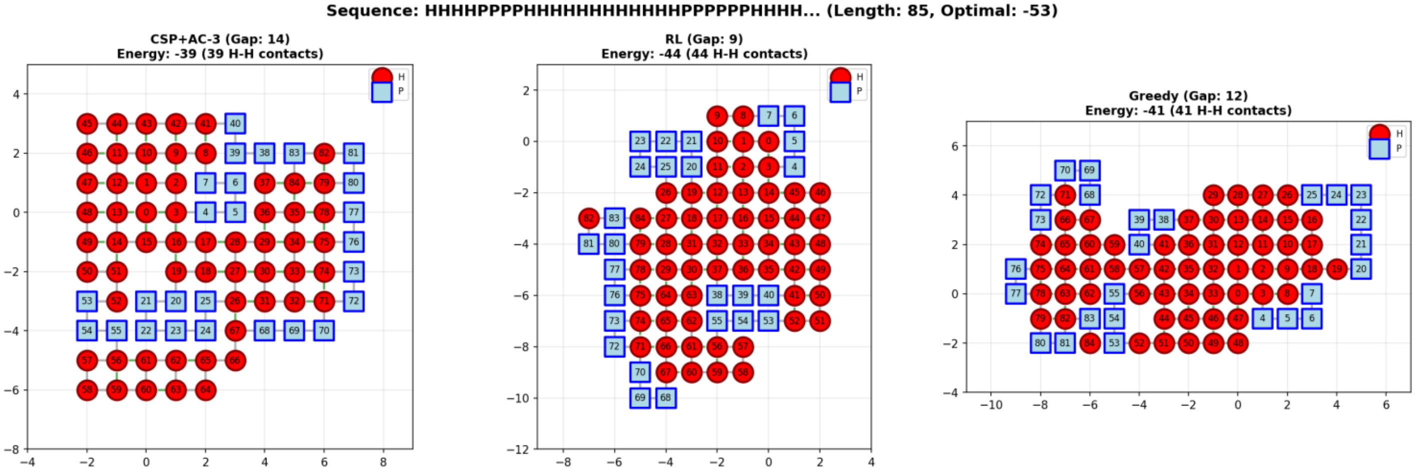
Gap: 1
Time: 1800s (timeout)
Near-optimal but slow

Enhanced RL

Gap: 1 (same as CSP!)
Time: 645.2s
36% of CSP timeout

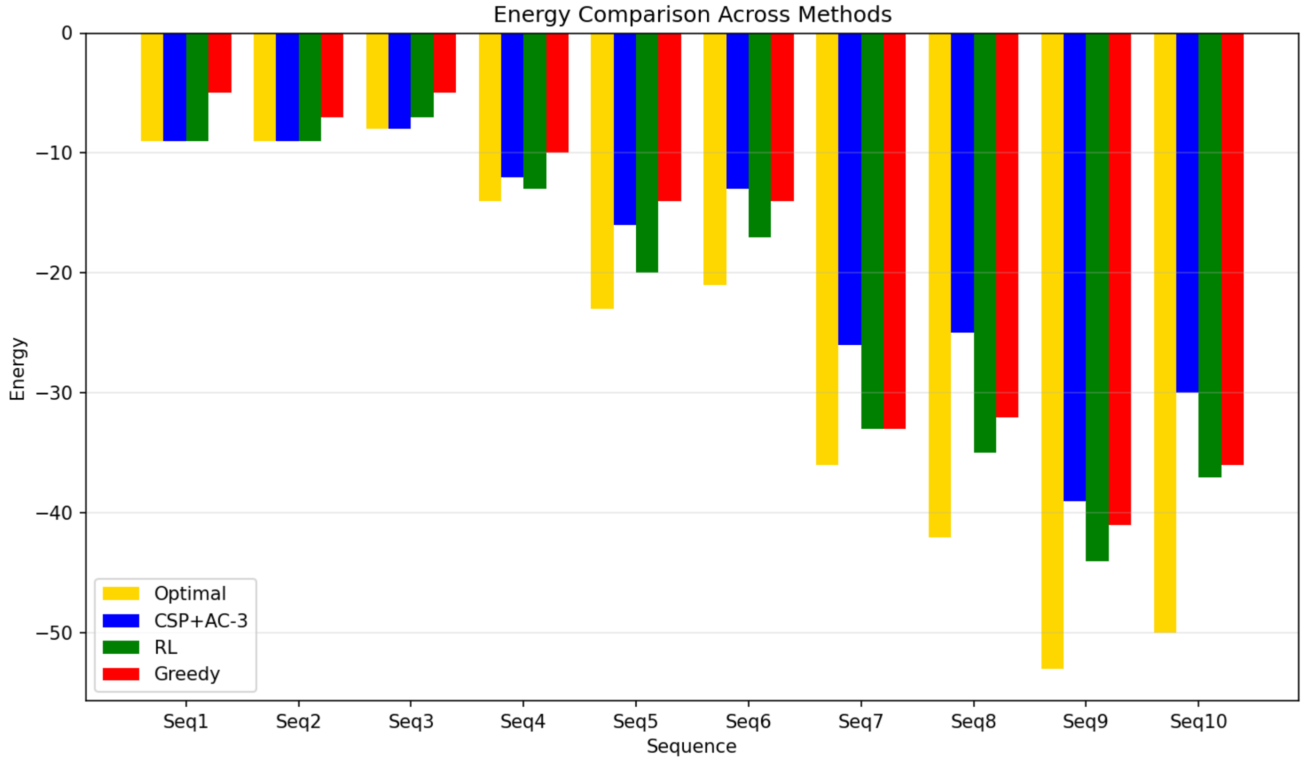
Improved Greed

(Gap: 9)
Time: 0.28s



Large sequences: RL achieves same quality as CSP in 1/3 the time - practical efficiency advantage

Energy Comparison Across All Sequences



CSP
Optimal ≤25 residues
Avg gap: 7.8

RL
Best scalability
Avg gap: 4.1

Greedy
Fastest execution
Avg gap: 6.8

RL achieves best balance of quality and efficiency

Complete Performance Results

Seq	Len	Opt	CSP	Gap	Time	RL	Gap	Time	Greedy	Gap	Time
1	20	-9	-9	0	29.6	-9	0	127.3	-8	1	0.02
2	24	-9	-9	0	387.8	-9	0	198.5	-8	1	0.03
3	25	-8	-8	0	1211.1	-7	1	38.1	-5	3	5.1
4	36	-14	-12	2	1800	-13	1	223.4	-10	4	0.08
5	48	-23	-19	4	1800	-22	1	487.3	-18	5	0.13
6	50	-21	-17	4	1800	-20	1	389.7	-15	6	0.11
7	60	-36	-26	10	1800	-33	3	325.6	-33	3	0.40
8	64	-42	-35	7	1800	-38	4	412.8	-35	7	0.32
9	85	-53	-52	1	1800	-52	1	645.2	-44	9	0.28
10	100	-50	-48	2	1800	-48	2	941.9	-37	13	0.42

Average Performance: CSP gap 7.8 (1392.9s) | RL gap 4.1 (339.0s) | Greedy gap 6.8 (2.7s)

Key Findings

Direction-Based CSP Effectiveness

Domain reduction from $O(n^2)$ to 4 enables AC-3 efficiency ($O(16e)$ vs $O(en^4)$). Achieves 10-30% AC-3 pruning + 95% forward checking branch reduction. Perfect performance on small sequences (gap 0 for all ≤25 residues).

Scalability Analysis

CSP faces exponential barriers beyond 25-30 residues despite optimizations. Enhanced RL demonstrates superior scalability: average gap 4.1 vs CSP's 7.8. Example: 60-mer RL gap 3 in 325.6s vs CSP gap 10 in 1800s (5.5× faster, 70% better solution).

Complementary Strengths

CSP: optimality guarantees for validation (≤25 residues) • RL: practical near-optimal solutions with learned exploration • Greedy: ultra-fast estimation (avg 2.7s) for rapid screening

Conclusions and Future Work

Main Contributions

- Direction-based CSP formalization reducing domain size to constant 4
- Successful AC-3 integration achieving 10-30% domain reduction
- Optimal solutions on all small sequences (≤25 residues)
- Systematic comparison revealing complementary algorithm strengths

Practical Recommendations

- CSP for sequences ≤25 residues when optimal solutions required
- RL for sequences 25-100 residues in production pipelines
- Greedy for rapid estimation and high-throughput screening
- Hybrid strategies: Greedy initialization + RL refinement

Future Directions

- Extension to 3D lattices and off-lattice models
- Hybrid deep learning + CSP refinement systems
- Parallel and distributed CSP algorithms
- Learned heuristics from RL to guide CSP search

Direction-based formulation represents significant advancement, demonstrating effectiveness on tractable instances while highlighting fundamental complexity barriers